ANALYSIS I — SHEET 2 (for Week 4 tutorials)
Suprema, Infima, Countability

1. For each of the following sets, write down *without proof* the (i) supremum; (ii) infimum; (iii) maximum; (iv) minimum or simply state that it is not defined.

$$\mathbb{Q} \cap [0, \sqrt{2}]; \qquad \{(-1)^n + 1/n \mid n \geqslant 1\}; \qquad \{3^n \mid n \in \mathbb{Z}\}; \qquad \bigcup_{n=1}^{\infty} \left[\frac{1}{2n}, \frac{1}{2n-1}\right].$$

2. By adapting the proof of the Archimedean Property, prove that $\{2^n \mid n \in \mathbb{N}\}$ is not bounded above. [*You may not make any use of logarithms.*]

3. Let S, T be non-empty bounded subsets of $\mathbb{R}$. Prove that:

(i) $S \cup T$ is bounded above, and $\sup(S \cup T) = \max(\sup S, \sup T)$;

(ii) $S + T = \{s + t \mid s \in S, t \in T\}$ is bounded below, and $\inf(S + T) = \inf S + \inf T$.

4. Prove that there exists a unique real number α such that $\alpha^3 = 2$. [*Hint: adapt the proof of the existence of* $\sqrt{2}$ *from lectures.*]

5. (i) Prove that there are uncountably many irrationals.

(ii) Let a, b be real numbers with $a < b$. Show that there is a natural number n such that

$$\frac{1}{n} < b - a.$$

Deduce that there is a rational number in the interval (a, b).

(iii) Show further that between any two real numbers there is an irrational number.

6. A complex number is said to be *algebraic* if it is the root of a polynomial with integer coefficients and otherwise is said to be *transcendental*.

(i) Show that there are countably many cubic polynomials with integer coefficients.

(ii) Show that there are countably many complex numbers which are roots of some integer-coefficient cubic polynomial.

(iii) Show further that there are countably many algebraic numbers and uncountably many transcendental numbers.

Extension Exercises (Optional)

7. (i) Let $a > 0$ and $q = m/n$ be rational with $n > 0$. Assuming that nth roots exist and are unique, show that rational powers are well-defined by

$$a^q = \sqrt[n]{a^m}.$$

(ii) Given rational numbers q_1, q_2, show that

$$a^{q_1+q_2} = a^{q_1}a^{q_2} \qquad \text{and} \qquad a^{q_1q_2} = (a^{q_1})^{q_2}.$$

8. Which of the following sets are countable? Justify your answers.
(i) The set of all subsets of $\mathbb{N}$.
(ii) The set of all finite subsets of $\mathbb{N}$.
(iii) The set of polynomials with integer coefficients.
(iv) The set of all rational sequences.

9. Construct a bijection between $[0, 1]$ and $[0, 1)$.

10. (i) Let A and B be non-empty bounded sets such that $a \leqslant b$ for all $a \in A$ and $b \in B$. Show that there exists c such that $a \leqslant c \leqslant b$ for all $a \in A$ and $b \in B$.

(ii) Show that (i) is equivalent to the completeness axiom. That is, assuming the axioms of an ordered field and the result in (i), deduce the completeness axiom.

11. (i) Let $x \in \mathbb{R}$. Show that there exists a unique integer n such that $n \leqslant x < n + 1$. We write $n = \lfloor x \rfloor$ and refer to n as the *floor* of x.

(ii) Likewise show that there exists a unique integer m such that $m - 1 < x \leqslant m$. We write $m = \lceil x \rceil$ and refer to m as the *ceiling* of x.

12. A train moves along the real line with constant integer velocity and starts out at $t = 0$ at an integer. At each time $t = 0, 1, 2, \ldots$ you are able to check whether the train is at an integer of your choosing (which may be a different choice each time). Explain how it is possible to determine the train's velocity and starting position.